

Robust Reversible Watermarking in Encrypted Image With Secure Multi-Party Based on Lightweight Cryptography

Lizhi Xiong¹, Xiao Han, Ching-Nung Yang², *Senior Member, IEEE*, and Yun-Qing Shi, *Life Fellow, IEEE*

Abstract—With the rapid development of network media, increasing research on reversible watermarking has focused on improving its robustness to resisting attacks during digital media transmission. There are some other reversible watermarking schemes that work in the encrypted domain for preserving the privacy of the cover image. The robustness of the watermarking and the privacy preserving of the cover image have become the key factors of reversible watermarking. However, there are few robust reversible watermarking schemes in the encrypted domain that could resist common attacks (such as JPEG compression, noise addition) and preserve privacy at the same time. In addition, the embedding capacity of a robust watermark and the efficiency of the encryption method must be considered. Recently, cloud computing technology has led to the rapid growth of network media, and many multimedia properties are owned by multiple parties, such as a film's producer and multiple distributors. Multi-party watermarking has become an important demand for network media to protect all parties' rights. In this paper, a Robust Reversible Watermarking scheme in Encrypted Image with Secure Multi-party (RRWEI-SM) based on lightweight cryptography is first proposed. Additive secret sharing and block-level scrambling are developed to generate the encrypted image. Then, the robust reversible watermarking based on significant bit Prediction Error Expansion (PEE) is performed by Secure Multi-party Computation (SMC). For applications with high robustness, a Modified RRWEI-SM is proposed by exploiting two-stage architecture. Furthermore, both the RRWEI-SM scheme and Modified RRWEI-SM scheme are

separable and can be applied to multiparty copyright protection. The experimental results and theoretical analysis demonstrate here that the RRWEI-SM and the Modified RRWEI-SM are secure, robust and effective.

Index Terms—Robust reversible Watermarking, secret sharing, encrypted image, lightweight cryptography.

I. INTRODUCTION

DIGITAL watermarking is widely utilized because of its integrity of information, anti-counterfeiting traceability and copyright protection. However, traditional watermarking techniques could bring irreversible distortion to the cover image. To retrieve the embedded message and recover the cover image, Reversible Watermarking (RW) has been proposed, and it has drawn increasing attention [1].

There are many methods that have been proposed to provide reversibility of the cover image in the watermarking scheme, e.g., lossless compression [2], [3], histogram shifting [4]–[9], difference expansion [10]–[12] and prediction error expansion [13]–[18]. In these methods, the cover image can be recovered from the marked image. However, such schemes are fragile to noise attacks and image compression. To resolve this issue, Robust Reversible Watermarking (RRW) is proposed.

In an RRW scheme, the watermark can be extracted and the cover image can be recovered when there is no distortion on the marked image. The distorted marked image caused by a malicious attack cannot be recovered to the original cover image, while the watermark is still expected to be correctly extracted [19], [20]. Many RRW schemes [21]–[27] have been proposed in recent years, most of which provide robustness of the watermark by developing traditional reversible techniques. These schemes can be classified into two categories: 1) Embedding the reversible watermark into the image's domain with a robust property (e.g., the Higher Significant Bits (HSBs) plane [24], [25] and high frequency domain). Thus, the robustness of the watermark is achieved by the property of certain domains. 2) Exploiting two-stage architecture for RRW, this was proposed by Coltuc *et al.* [21], [22]. In the second category, a secret message is embedded based on robust watermarking in the first stage. Then, the difference between the marked image and the original cover image is recorded. The difference information is embedded into the marked image in the second stage, and it can be used to recover the original cover image. However, when the payload of this scheme is increased, the visual quality of the marked image will be reduced significantly. Thus,

Manuscript received June 6, 2020; revised September 24, 2020; accepted January 12, 2021. Date of publication February 1, 2021; date of current version January 11, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61702276 and Grant 61672294, in part by the Ministry of Science and Technology (MOST), under Grant MOST 108-2221-E-259-009-MY2, in part by the Startup Foundation for Introducing Talent of Nanjing University of Information Science and Technology under Grant 2016r055, and in part by the Priority Academic Program Development (PAPD) of Jiangsu Higher Education Institutions. This article was recommended by Associate Editor P. Bestagini. (Corresponding authors: Lizhi Xiong; Ching-Nung Yang.)

Lizhi Xiong is with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing 210044, China, and also with the Engineering Research Center of Digital Forensics, Ministry of Education, Jiangsu Collaborative Innovation Center on Atmospheric Environment and Equipment Technology, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: lzxiong16@163.com).

Xiao Han is with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: hanxiao6207@qq.com).

Ching-Nung Yang is with the Department of Computer Science and Information Engineering, National Dong Hwa University, Hualien 974, Taiwan (e-mail: cnyang@gms.ndhu.edu.tw).

Yun-Qing Shi is with the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ 07102 USA.

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2021.3055072>.

Digital Object Identifier 10.1109/TCSVT.2021.3055072

1051-8215 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

the embedding capacity of these RRW schemes are limited. Moreover, the abovementioned schemes work in the plaintext domain, which means that the privacy of the cover image is not well protected. In many applications scenarios, e.g., outsourcing services, cloud computing platforms and multi-party copyright protection [28], [29], the privacy preserving feature is crucial.

To preserve the privacy of the cover image, many reversible data hiding schemes in the encrypted domain have been proposed [17], [18], [30]–[33]. Three different roles, i.e., image provider, data hider, and receiver, have been involved in RDHEI. To ensure the correctness of the RDHEI, public key and homomorphic encryption-based schemes [17], [33] are proposed. However, the computational complexity of these schemes is relatively high. Inspired by the schemes that utilize the homomorphic property of the encryption, secret sharing-based RDHEI has been proposed [8], [32]. However, in many applications of network media, multimedia properties are owned by multiple parties. More than one participant must be involved. The existing RDHEI schemes enable only one participant to embed the secret message. Thus, these schemes are not suitable for multi-party copyright protection. Most importantly, these schemes are not robust.

Let us briefly summarize the abovementioned issues: 1) The embedding capacity of traditional RRW is limited; 2) The RRW in the plaintext domain results in the leakage of privacy; 3) The complexity of the encryption method for the cover image is relatively high; 4) The inseparable schemes would reduce the practicality of robust reversible watermarking; 5) Current RRW schemes are not suitable for multiparty copyright protection applications.

To solve the abovementioned issues, we propose a new method that we call Robust Reversible Watermarking in Encrypted Image with Secure Multi-party (RRWEI-SM), which is based on lightweight cryptography. Inspired by the Secure Multi-party Computation (SMC) in the semi-honest (also known as passive or honest-but-curious) security model [34], additive secret sharing is developed as a lightweight encryption method in our scheme. In this scheme, the order of the extraction of the watermark, the recovery of the cover image and the decryption of the encrypted image are commutative, i.e., the RRWEI-SM is separable. Based on the RRWEI-SM, a Modified RRWEI-SM is also proposed by exploiting the two-stage architecture. The Modified RRWEI-SM has better robustness at the low embedding rate.

The remainder of this paper is organized as follows. Section II introduces the related work. The preliminaries are described in Section III. The proposed schemes are presented in Section IV. Experiments and discussion are shown in Section V. The conclusions are presented in Section VI.

II. RELATED WORK

In this section, related work (including robust reversible watermarking and reversible data hiding in encrypted images) is described in the upcoming subsections, respectively. Finally, our contributions are also drawn.

A. Robust Reversible Watermarking

Currently, the research on RRW has developed reversible watermarking techniques that provide robustness and

reversibility for the schemes. They are mainly based on two-stage architecture or on an image domain with robust features.

A two-stage architecture for RRW was proposed by Coltuc *et al.* [21], [22]. The robust watermark is embedded into the cover image in the first stage to generate an intermediate image. In the second stage, namely, the reversible embedding stage, the information for recovery is embedded into the intermediate image by the reversible watermarking method. Because the reversible watermark is embedded into the domain that was used in the first stage, there exists an interference in the robust watermark. In [23], Wang *et al.* proposed a method by using an independent embedding domain to address the problem. This scheme exploited the traditional patchwork robust watermarking technique in the first stage (i.e., the robust embedding stage), and it enhanced the performance of the two-stage RRW. However, the embedding capacity of the traditional robust watermarking scheme is limited.

In addition, there is another way to construct robust reversible watermarking. Some traditional reversible watermarking methods can be modified to achieve robustness [24]–[27]. Wang *et al.* [24] proposed a lossless robust data hiding scheme based on Significant-Bit-Difference Expansion (SBDE). In [24], the pixels in the cover image are first decomposed into two parts: the Higher Significant Bits (HSBs) and the Least Significant Bits (LSBs). Then, the HSB differences between the adjacent pixels are calculated. Secret bits are embedded into HSB by shifting the histogram of the difference value. Kumar and Jung [25] improved the work of Wang *et al.* [24] by exploiting a two-layer embedding strategy and making a Prediction Error Expansion (PEE) on significant bits. The visual quality of the marked image is improved to some degree. In these schemes, nonmalicious attacks, such as JPEG compression, will not cause disturbances to the main content of the image as well as the hidden secret data. However, in these schemes, the watermark is embedded into the plaintext domain, where the privacy of the cover image is not preserved.

B. Reversible Data Hiding in Encrypted Images

Many RDHEI schemes have been proposed for applications that require privacy protection, where the watermark is embedded into the encrypted image. The state-of-the-art research focuses on improving the embedding capacity and the security of the encrypted image. In the separable schemes, the visual quality of the marked image must also be guaranteed. In [30], Most Significant Bits (MSBs) prediction is used to recover the cover image. Then, the watermark is embedded into the MSBs of the encrypted image. However, the scheme is inseparable. Ke *et al.* [17] exploited learning with error-based fully homomorphic encryption to preserve the privacy of the cover image. In this scheme, a secret message is embedded into the encrypted image by difference expansion. In addition, key-switching and bootstrapping are introduced in this scheme to control a ciphertext extension and make the scheme extendable for other data hiding approaches. In [18], somewhat homomorphic encryption is exploited to generate the encrypted image, in which the Prediction Error

TABLE I
ADVANTAGES AND DISADVANTAGES OF EXISTING WORKS

Method	Robustness	Privacy preserving	Decryption complexity	Separable	Multi-party watermarking
Wu <i>et al.</i> [8]	No	Yes	$O(n \log^2 n)$	Yes	No
Ke <i>et al.</i> [17]	No	Yes	$O(n^2)$	Yes	No
Xiong <i>et al.</i> [18]	No	Yes	$O(n^2)$	Yes	No
Coltuc and Chassery [22]	High	No	-	-	No
Wang <i>et al.</i> [23]	High	No	-	-	No
Wang <i>et al.</i> [24]	Low	No	-	-	No
Kumar and Jung [25]	Low	No	-	-	No
Puteaux and Puech [30]	No	Yes	$O(n)$	No	No
Xiang <i>et al.</i> [31]	No	Yes	$O(n^3)$	Yes	No
Chen <i>et al.</i> [32]	No	Yes	$O(n \log^2 n)$	No	No

Expansion (PEE) method is performed to embed the secret data. However, the computational complexity of such public key-based schemes are not low.

To reduce the computational complexity of the encryption, Wu *et al.* [8] proposed a RDHEI that is based on secret sharing. The cover image is encrypted into more than one noise-like share, and the secret message is embedded into these shares based on difference expansion or histogram shifting. Since the cover image is encrypted by polynomial-based secret sharing, the encrypted shares can be decrypted according to a Lagrange interpolation operation. Chen *et al.* [32] also developed polynomial-based secret sharing to propose a reversible data hiding scheme in encrypted images using lightweight cryptographic algorithms. First, the coefficients of a 4-degree polynomial are substituted by two pixels' values from the cover image and two random integers. Then, two shares are generated by the polynomial. By utilizing the inherently additive homomorphism of the polynomial, the secret data can be embedded into the shares based on difference expansion. The scheme in [32] achieves good visual quality of the marked image, and the maximum payload of this scheme is 0.5 *bpp*. However, such schemes are not extendable for multi-party copyright protection. Most importantly, these RDHEI schemes are not robust.

Generally, the advantages and disadvantages of existing works are listed in Table I. The schemes in [22]–[25] have robustness and reversibility, but the privacy of the cover image cannot be preserved. Thus, the decryption complexity and the separable property is not considered. In [8], [17], [18] [30]–[32], the cover image is firstly encrypted, where the data hider cannot obtain any information from the encrypted image. However, the embedded data is not robust. Moreover, computational complexity of the scheme in [17] is $O(n^2)$, and that is relatively high for some applications. In addition, the maximum payload of the schemes in [23] and [31] are 0.25 *bpp* and 0.384 *bpp* [32], respectively. Thus, these schemes cannot be applied to the applications with high payload. The scheme in [30] achieves a payload of 1 *bpp*, but the scheme is not separable. Most importantly, all of

these schemes are not suitable for multi-party watermarking applications.

Therefore, a robust reversible watermarking in encrypted image with high payload, low computational complexity, and suitable for a multi-party watermarking application is needed.

C. Contributions of This Paper

This paper proposes Robust Reversible Watermarking in Encrypted Image with Secure Multi-party (RRWEI-SM), which is based on lightweight cryptography, and it includes RRWEI-SM and Modified RRWEI-SM. Overall, the proposed schemes are secure, robust, effective and suitable for multiparty copyright protection applications. Compared with existing schemes, our contributions are as follows:

1. We propose a lightweight image encryption method that is based on additive secret sharing and block-level scrambling. The watermark is embedded into the encrypted shares based on Secure Multi-party Computation (SMC).
2. Based on lightweight image encryption, we propose the Robust Reversible Watermarking in Encrypted Image with Secure Multi-party (RRWEI-SM), which can preserve the privacy of the cover image and ensure the robustness of the watermark. Furthermore, the RRWEI-SM scheme has a higher embedding capacity and a better visual quality compared with many reversible watermarking schemes in encrypted domain and robust watermarking schemes in the plaintext domain.
3. We further propose the Modified RRWEI-SM scheme, which exploits the two-stage architecture. The patchwork robust watermark and PEE-based reversible watermark are embedded into the shares based on SMC. The Modified RRWEI-SM has a stronger robustness compared with many other RRW schemes. Moreover, the RRWEI-SM scheme and the Modified RRWEI-SM scheme are suitable for multiparty copyright protection applications.

III. PRELIMINARIES

In this section, we present the concepts that inspired us when creating the RRWEI-SM scheme. Wang *et al.*'s

Significant-Bit-Difference Expansion (SBDE) scheme is first introduced. Then, the block-level predictor of the PEE is formulated. The concept of the Secure Multi-party Computation (SMC) is described last.

A. Wang *et al.*'s SBDE Scheme

Wang *et al.* [24] proposed the Significant Bit Difference Expansion scheme (SBDE) to provide robustness for reversible data hiding. A grayscale image (size $M \times N$) is used as a cover image, in which the pixel values vary in the range 0-255. Each pixel ($P_{i,j}$) of the cover image can be represented by 8 bits in binary form. It has been observed that some of the unintentional modification attacks, such as JPEG compression, have an impact on only the LSBs of the cover image. Accordingly, the SBDE scheme decomposes the pixels of the cover image into the Higher Significant Bits (HSBs) plane and the Least Significant Bits (LSBs) plane. Additionally, the HSB plane is utilized to provide robustness for the scheme. The decomposition method is as follows:

$$P_{i,j} = \sum_{k=n}^7 b_k \times 2^k + \sum_{k=0}^{n-1} b_k \times 2^k \quad (1)$$

where

$$h_{i,j} = \sum_{k=n}^7 b_k \times 2^k \quad (2)$$

$$l_{i,j} = \sum_{k=0}^{n-1} b_k \times 2^k \quad (3)$$

Here, $h_{i,j}$ represents the HSB plane of pixel $P_{i,j}$, $l_{i,j}$ represents the LSB plane of pixel $P_{i,j}$, and $b_k \in \{0, 1\}$ denotes the bit value at the k -th location of the 8 bits, whose value is dependent on the value of n , which can vary from 1 to 8. The value of $h_{i,j}$ ranges from 0 to $2^{(8-n)} - 1$. The value of $l_{i,j}$ varies in the range 0 to $2^n - 1$.

The SBDE then uses a difference expansion method to embed the secret bits into I_{HSB} . First, the difference between two consecutive a pixel rowwise is calculated as follows:

$$d_{i,j} = h_{i,j} - h_{i,j-1} \quad (4)$$

where $i \in [1, M]$ and $j \in [2, N]$. The differences between the pixels in the first column of the image are calculated by

$$d_{i,1} = h_{i,1} - h_{i-1,1} \quad (5)$$

where $i \in [2, M]$.

Thus, the histogram of the differences between the adjacent pixels can be constructed for the embedding process. We assume that a_1 and a_2 ($a_1 < a_2$) are the histogram bins with the maximum and the second maximum difference-value intensity, respectively. The bins (B) such as $B < a_1$ or $B > a_2$ are shifted to vacate room for embedding secret bits, and the remaining bins for $a_1 < B < a_2$ remain the same. Last, the secret bits are embedded into the bins where $B = a_1$ and $B = a_2$. The HSB plane of the image has robustness against minor modification attacks such as JPEG. In addition, the high correlation between the pixels in I_{HSB} provides a high embedding capacity.

B. Block-Level Predictor (BLP)

The BLP is a predictor based on $n \times n$ image blocks. We take the 2×2 block as an example. It enables the target pixel value $x(i, j)$, where $i \in [1, M]$, $j \in [1, N]$, to be predicted by the remaining pixels with weights in the block. The calculation process is as follows:

$$\hat{x} = \lfloor w_r x_r + w_c x_c + w_d x_d \rfloor \quad (6)$$

where (x, x_r, x_c, x_d) represents the pixels in the same block, and (w_r, w_c, w_d) denotes the weight coefficients. They are satisfied with $0 \leq w_r, w_c, w_d \leq 1$, $w_r + w_c + w_d = 1$.

For example, when three identical coefficients are used in PEE, the target pixel $x(i, j)$ can be predicted by the average values of the remaining three pixels. After the prediction, the PEE can be performed to embed the secret bits.

C. Secure Multi-Party Computation (SMC)

Secure Multi-party Computation (SMC) is an interactive computational paradigm. Multiple parties collaborate to operate a function over their respective inputs while the privacy of the inputs is preserved.

Many SMC schemes [34], [37] have made use of secret sharing techniques that were introduced by Shamir [35] and Blakey [36]. Additive secret sharing and Shamir secret sharing are commonly used in these schemes because of their efficiency and because they do not rely on any computationally expensive cryptographic primitives. In SMC, the related operations are under the honest-but-curious security model. In [34], the security of additive secret sharing has been proven, which means that the privacy is well preserved.

IV. THE PROPOSED SCHEME

Here, the design concept of the proposed schemes (the RRWEI-SM and the Modified RRWEI-SM) is first introduced in Section IV-A. Then, the different phases of the RRWEI-SM scheme is shown in Section IV-B. The Modified RRWEI-SM scheme is presented in Section IV-C. Finally, theoretical analysis is given.

A. Design Concept

To solve the issues presented above in Section I, we propose the Robust Reversible Watermarking in Encrypted Image with Secure Multi-party (RRWEI-SM), which is based on lightweight cryptography. In the RRWEI-SM, additive secret sharing and block-level scrambling are developed as a lightweight encryption method that preserves the privacy of the cover image. The encrypted shares are obtained by the encryption method and are distributed to different parties. In the embedding phase, the performance of PEE is often better than that of DE, and therefore, the PEE on significant bits is applied in our scheme, not SBDE. In this way, we first exploit the PEE on significant bits to embed the robust watermarking into the HSB plane of the shares, and this approach achieves a high embedding capacity. The embedding operation is performed based on SMC, and it would not cause a privacy leakage. Moreover, the decryption and extraction phases are separable.

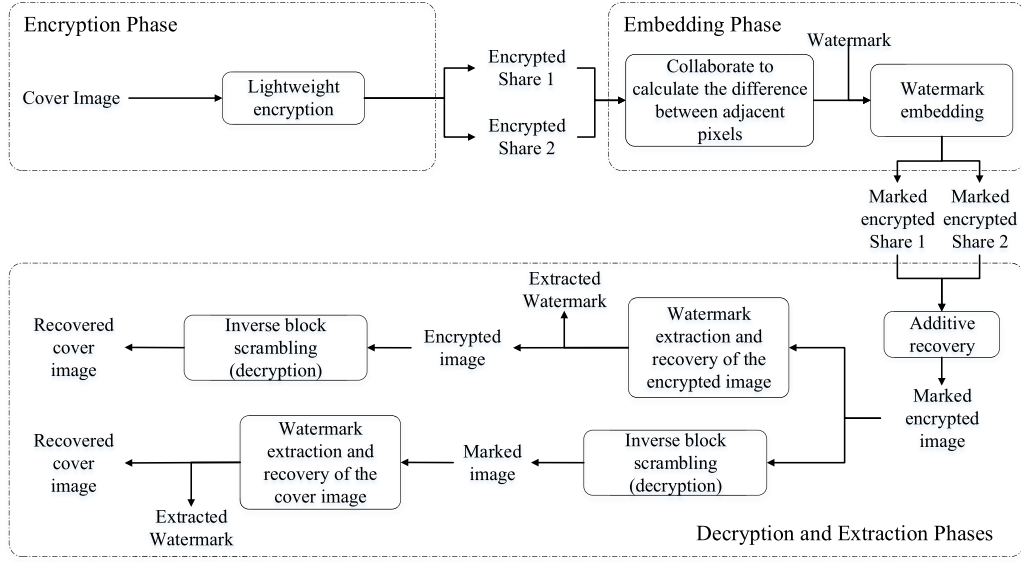


Fig. 1. The framework of the RRWEI-SM.

For applications with high robustness, a Modified RRWEI-SM is proposed. A two-stage architecture is introduced to the RRWEI-SM. The patchwork robust watermarking and the Prediction-Error Expansion (PEE) reversible watermarking are utilized. In the Modified RRWEI-SM, the extraction of the watermark and the decryption of the cover image can also be completed separable.

The framework of the RRWEI-SM (including the Modified RRWEI-SM) is shown in Fig. 1. First, a cover image is encrypted by a lightweight image encryption method based on additive secret sharing and block-level scrambling. For simplicity, there are two parties in the framework. Then, two encrypted shares are obtained, and they are distributed to the two parties. In the embedding phase, the two parties collaborate according to the secure addition protocol to embed the watermark into their owned shares. The two marked encrypted shares can be used to obtain the marked encrypted image by an addition operation. Finally, the user can choose to directly decrypt the encrypted image and obtain the marked image, which is similar to the original image.

Then, extraction of the watermark and recovery of the image are completed. It is remarkable that the order of the extraction and the decryption is commutative.

B. The RRWEI-SM Scheme

The RRWEI-SM scheme consists of three phases: image encryption, watermark embedding, decryption and extraction phases. The different phases and their examples are shown in this section.

1) *Image Encryption Phase:* To preserve the privacy of the cover image, we introduce a lightweight image encryption that is based on additive secret sharing and block-level scrambling. For simplicity, additive secret sharing splits the pixel value of the cover image (I is size $M \times N$) into two positive integers, randomly:

$$x(i, j) = x_1(i, j) + x_2(i, j) \quad (7)$$

where $x(i, j)$ represents the pixel value of the cover image, $i \in [1, M]$, $j \in [1, N]$. Here, $x_1(i, j)$ and $x_2(i, j)$ denote the pixel values of the shares I_1 and I_2 , respectively. To reduce the correlation between the pixels, we introduce a block-level scrambling process to enhance the security of the encryption method. The cover image is divided into nonoverlapping blocks, and then, these blocks are scrambled. The size of the block is determined by the prediction method. For example, if a 2×2 block is used to predict the pixel value in the watermark embedding phase, the 2×2 block scrambling is performed. There is no limit to the specific method of scrambling in the proposed scheme, and here, we take the high-speed-scrambling method as an example. It has advantages in many respects, i.e., small key space, fast speed and high security [38].

It is remarkable that only the HSB plane of the image shares is used to calculate the prediction error. As a result, the additive secret sharing method must be modified to ensure the correctness. First, the pixels of the cover image must be decomposed into n LSBs and $(8-n)$ HSB:

$$x(i, j) = x_{HSB}(i, j) \times 2^n + x_{LSB}(i, j) \quad (8)$$

where

$$x_{HSB}(i, j) = \frac{x(i, j) - \text{mod}(x(i, j), 2^n)}{2^n} \quad (9)$$

$$x_{LSB}(i, j) = \text{mod}(x(i, j), 2^n) \quad (10)$$

Here, $x_{HSB}(i, j)$ represents the value of the HSB, and $x_{LSB}(i, j)$ is the value of the LSB. Then, $x_{HSB}(i, j)$ and $x_{LSB}(i, j)$ are randomly split by the addition operation.

$$x_{HSB}(i, j) = x_{HSB}^1(i, j) + x_{HSB}^2(i, j) \quad (11)$$

$$x_{LSB}(i, j) = x_{LSB}^1(i, j) + x_{LSB}^2(i, j) \quad (12)$$

Next, $x_1(i, j)$ and $x_2(i, j)$ are obtained by

$$x_1(i, j) = x_{HSB}^1(i, j) \times 2^n + x_{LSB}^1(i, j) \quad (13)$$

$$x_2(i, j) = x_{HSB}^2(i, j) \times 2^n + x_{LSB}^2(i, j) \quad (14)$$

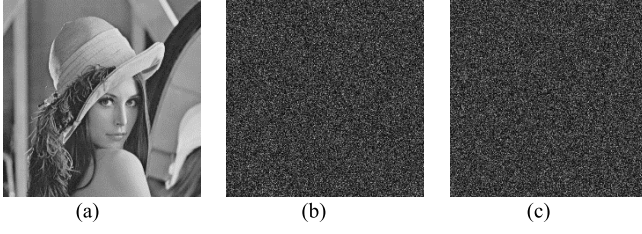


Fig. 2. (a) cover image "Lena" (b) the encrypted shares 1 (c) the encrypted shares 2.

The two parts of the pixel value are uniformly distributed; thus, the property of additive secret sharing is preserved. If $x(i, j)$ is split by addition directly, then the differences calculated by HSB of the two shares' pixels might not equal the differences between the pixels of the original image. That will cause an error in the prediction error calculation.

Last, the encrypted shares 1 and 2 are distributed to two edge servers in the honest-but-curious model. The results of the different stages are shown in Fig. 2.

We choose 4 blocks, as shown in Fig. 3, to demonstrate the encryption phase. First, these blocks are scrambled. Then, each block is shared by additive secret sharing, and the two encrypted shares are obtained. An example of additive secret sharing is shown in Fig. 3. The pixel value '165' is first divided into two parts: the HSB plane (20×2^3) and the LSB plane (5). Then, these two planes are divided by additive secret sharing randomly. We can see that the pixel values in the shares are independent from the pixel values in the original cover image.

2) **Watermark Embedding Phase:** The significant bits PEE method based on SMC is utilized in the watermark embedding phase. We take the 2×2 block-level predictor as an example. Let $\widehat{x}_{HSB}(i, j)$ represent the prediction value of $x_{HSB}(i, j)$, which is the average value of the remaining three pixels' HSB in the same block, (15), shown at the bottom of the next page, where (i, j) indicates the position of the upper left corner of the block.

The prediction error $e_{HSB}(i, j)$ can be calculated by

$$e_{HSB}(i, j) = x_{HSB}(i, j) - \widehat{x}_{HSB}(i, j) \quad (16)$$

In our scheme, to obtain $e_{HSB}(i, j)$ from the two shares, the two parties calculate part of $e_{HSB}(i, j)$, as follows:

$$e_{HSB}^1(i, j) = 3 \times x_{HSB}^1(i, j) - x_{HSB}^1(i, j+1) - x_{HSB}^1(i+1, j) - x_{HSB}^1(i+1, j+1) \quad (17)$$

$$e_{HSB}^2(i, j) = 3 \times x_{HSB}^2(i, j) - x_{HSB}^2(i, j+1) - x_{HSB}^2(i+1, j) - x_{HSB}^2(i+1, j+1) \quad (18)$$

When the embedding phase begins, the party holds the watermark and sends a request to another party for another part of $e_{HSB}(i, j)$.

$$e_{HSB}(i, j) = \left\lfloor \frac{e_{HSB}^1(i, j) + e_{HSB}^2(i, j)}{3} \right\rfloor \quad (19)$$

We can obtain the set of predictive errors represented as $E = \{e_{HSB}(i, j) | i \in [1, M], j \in [1, N]\}$. With the predictive errors E of the HSB of the cover image, the watermark can

be embedded into shares by PEE and histogram shifting.

$$x_k(i, j) = \begin{cases} x_k(i, j) + b \times 2^n, & \text{if } e_{HSB}(i, j) = M_E \\ x_k(i, j) + 2^n, & \text{if } e_{HSB}(i, j) > M_E \\ x_k(i, j), & \text{if } e_{HSB}(i, j) < M_E \end{cases} \quad (20)$$

where $b \in \{0, 1\}$ is one bit of the watermark, and $k \in \{1, 2\}$ represents the number of the share. M_E is defined as $M_E = \arg \max (h_E(e_{HSB}(i, j)))$, and $h_E(t)$ is the number of occurrences when the prediction-error values in the sequence E are equal to t .

The encrypted share 1 and the marked encrypted share 1 are shown in Fig. 4. Note that the value of encrypted share does not exceed the range from 0 to 255. The locations of the overflow/underflow pixels would be recorded, and the watermark will not be embedded into these pixels. Different parties embed their watermarks into independent areas of the shares.

When the watermark bits are embedded into share 1, the prediction-error in the corresponding block of share 2 is required for the owner of share 1. The owners of the shares calculate $e_{HSB}^1(i, j)$ and $e_{HSB}^2(i, j)$. In the example in Fig. 5, the owner of share 1 obtains the values '32' and '-32', and $e_{HSB}(i, j) = 0$ can be obtained by Eq. (19). Because M_E of the predictive error is equal to 0, one bit can be embedded into this block of share 1. After the embedding, the value of $x_1(i, j)$ is changed from 18 to 26. When 4 bits $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are embedded into the block of the share 1, the obtained marked of share 1 is shown in Fig. 6.

3) **Decryption and Extraction Phase:** The marked encrypted shares can be obtained from the two parties after the embedding phase. By adding the two marked encrypted shares, the marked encrypted image is revealed. Two cases are described as follows.

Watermark extraction and image recovery before decryption

Watermark extraction is an inverse process of watermark embedding. The marked encrypted image is first divided into blocks with the same size as that in the embedding phase. Then, the prediction error set E can be obtained by Eqs. (15) and (16). The watermark can be extracted by

$$b = \begin{cases} 0, & \text{if } e_{HSB}(i, j) = M_E \\ 1, & \text{if } e_{HSB}(i, j) = M_E + 1 \end{cases} \quad (21)$$

and the location with overflow will be skipped.

Then, the pixel values are recovered by

$$x(i, j) = \begin{cases} x(i, j) - 2^n, & \text{if } e_{HSB}(i, j) > M_E \\ x(i, j), & \text{if } e_{HSB}(i, j) \leq M_E \end{cases} \quad (22)$$

By this means, the watermark is extracted completely, and the encrypted image is also recovered. The original cover image can be obtained.

Watermark extraction and image recovery after decryption

With the marked encrypted image, the inverse scrambling can be completed before the extraction and recovery phases. The result of the decryption and extraction is shown in Fig. 7. After calculating the predictive error of the embedded pixels,

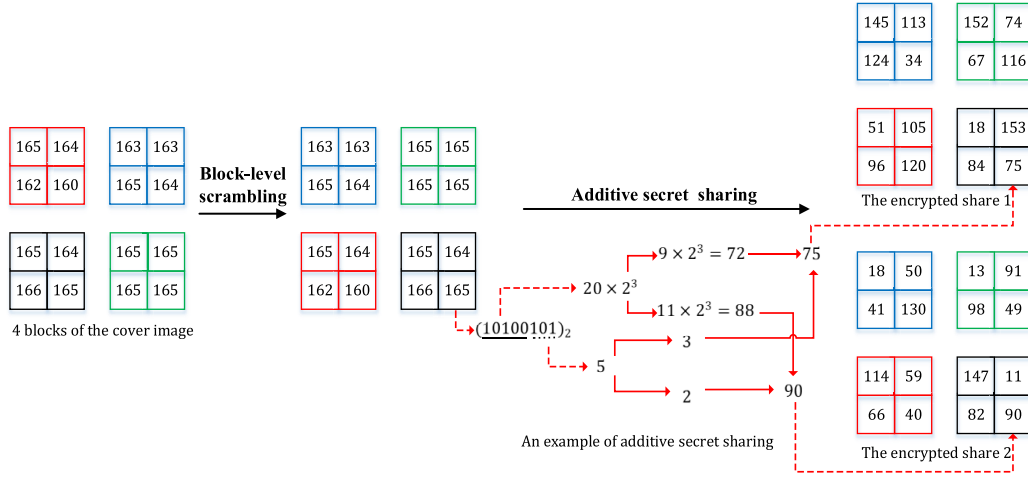


Fig. 3. An example of the encryption phase.

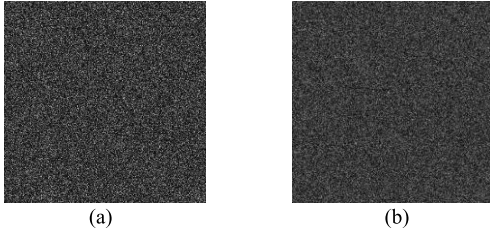


Fig. 4. (a) the encrypted share 1 (b) the marked encrypted share 1.

the watermark can be extracted by Eq. (21), and the marked image can be recovered by Eq. (22). Last, the extracted bits of the watermark are inversely scrambled.

As shown in Fig. 8, with the marked share 1 and the non-marked share 2, the receiver could add these shares together to obtain the encrypted image. In the first case, if the extraction by the PEE method is first conducted, the watermark bits can be extracted. The inverse block-level scrambling is completed in the last step. In another case, the receiver first decrypts the encrypted marked image by inverse block-level scrambling. After the decryption phase, the extraction and recovery can still be completed in each block of the marked image. The cover image will be recovered to be the same as the original image. In addition, the original watermark can be obtained by inverse block-level scrambling.

C. The Modified RRWEI-SM Scheme

To improve the robustness of the RRWEI-SM, we modify it by exploiting the two-stage RRW architecture [22]. In the first embedding stage, the patchwork robust watermarking is performed on the HSB plane of the shares. Then, the side information for recovering the cover image is embedded by the PEE method.

1) *Image Encryption Phase*: The image encryption method in the Modified RRWEI-SM is the same as the original method

in Section IV-B. For simplicity, the encryption and decryption phases will not be described repeatedly. The description of the embedding and extraction phases will focus on the processing of the shares.

2) *Watermark Embedding Phase*: In the Modified RRWEI-SM, the two-stage architecture [22] is exploited. In the first embedding stage, the patchwork robust watermark is embedded into the shares. The reversible watermarking for recovering the cover image based on PEE is performed in the second embedding stage. In addition, we can also compress the side information to reduce the bits which are embedded into cover image by LSB replacement [23].

First, the shares are divided into nonoverlapping blocks with $2m$ pixels $\{x_1, x_2, \dots, x_{2m}\}$. A random mapping is defined as $\{1, 2, \dots, 2m\} \Rightarrow \{\sigma(1), \sigma(2), \dots, \sigma(2m)\}$. Then, the difference d_i between two pixel sets in a block can be calculated as

$$d_i = \frac{\sum_{l=1}^m x_{\sigma(l)} - \sum_{l=m+1}^{2m} x_{\sigma(l)}}{m} \quad (23)$$

These difference values will be saved as side information used for recovering and will be embedded in the second stage by PEE method. Before that, one robust watermark bit $w \in \{0, 1\}$ is embedded into the block by

$$y_{\sigma(l)} = \begin{cases} x_{\sigma(l)} + (2w - 1) \times T - \frac{d_i}{2}, & \text{if } 1 \leq l \leq m \\ x_{\sigma(l)} - (2w - 1) \times T + \frac{d_i}{2}, & \text{if } m + 1 \leq l \leq 2m \end{cases} \quad (24)$$

where T is the threshold to control the robustness of the watermark. After embedding, the difference d_i is modified to

$$d_i^w = \frac{\sum_{l=1}^m y_{\sigma(l)} - \sum_{l=m+1}^{2m} y_{\sigma(l)}}{m} = \begin{cases} 2 \times T, & \text{if } w = 1 \\ -2 \times T, & \text{if } w = 0 \end{cases} \quad (25)$$

$$\widehat{x_{HSB}}(i, j) = \left\lfloor \frac{x_{HSB}(i, j+1) + x_{HSB}(i+1, j) + x_{HSB}(i+1, j+1)}{3} \right\rfloor \quad (15)$$

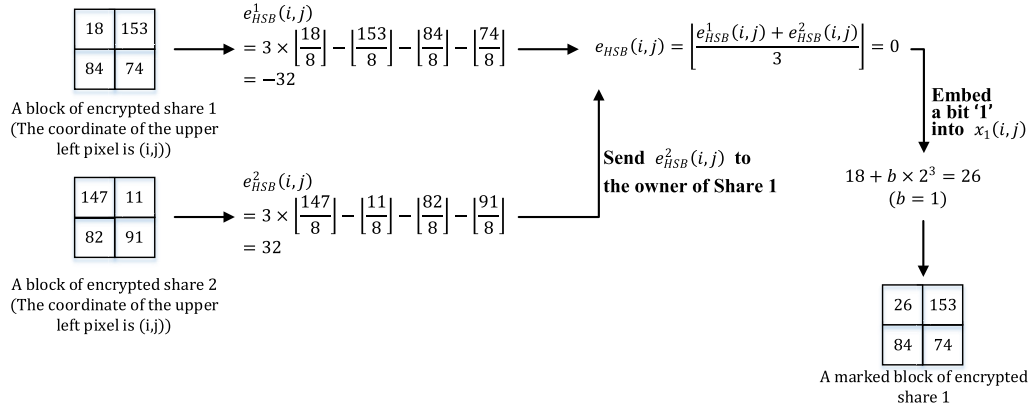


Fig. 5. An example of the embedding phase.

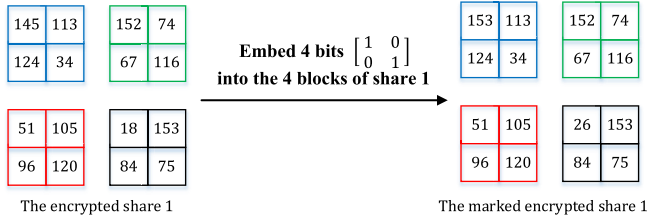


Fig. 6. The result of the embedding phase.



Fig. 7. (a) the marked encrypted image obtained by adding two shares (b) the marked image from direct decryption (c) the recovered cover image, which is exactly the same as the original cover image.

Notice that the size of the block also affects the robustness of the watermark. During embedding, if overflow/underflow problem occurs, the pixels in this block will be unchanged. The location of the overflow/underflow blocks will be recorded in the side information for recovering the cover image.

It should be noted that the patchwork robust watermark is embedded into the HSB plane of the shares, which enhances the robustness of the watermark. An example of the robust embedding phase is shown in Fig. 9.

3) *Decryption and Extraction Phase:* After combining the two shares by the addition operation, the patchwork robust watermark can be extracted by calculating the difference d_i^w between two pixel sets in a block. The rule of extraction is formulated as

$$w = \begin{cases} 1, & \text{if } d_i^w > 0 \\ 0, & \text{if } d_i^w \leq 0 \end{cases} \quad (26)$$

If the marked image is not attacked, the side information used for recovering the cover image can be extracted by the PEE method. The original difference values d_i recorded in the reversible watermark are used to recover the cover image, as

follows:

$$x_{\sigma(l)} = \begin{cases} y_{\sigma(l)} - (2w - 1) \times T + \frac{d_i}{2}, & \text{if } 1 \leq l \leq m \\ y_{\sigma(l)} + (2w - 1) \times T - \frac{d_i}{2}, & \text{if } m + 1 \leq l \leq 2m \end{cases} \quad (27)$$

An example of the extraction and recovery phase is shown in Fig. 10. The marked block is first obtained by adding the two shares together. Here, d_i^w is calculated from the HSB values of the marked pixels. The embedded robust watermark bit is extracted according to d_i^w . In addition, the original difference d_i can be extracted from the intact marked image, and it will be used to recover the pixel values as presented in Fig. 10.

Overall, both the RRWEI-SM and the Modified RRWEI-SM scheme can embed the robust reversible watermark into the encrypted image, and they are suitable for multi-party watermark applications. The RRWEI-SM embeds the reversible watermark into the robust domain of the cover image, i.e., the HSB plane. The Modified RRWEI-SM embeds the patchwork robust watermark into the robust domain of the cover image, and that further enhances the robustness of the watermark. This difference between the RRWEI-SM and the Modified RRWEI-SM scheme makes them suitable for different applications. Specifically, the visual quality of the marked image and the embedding capacities of the RRWEI-SM scheme are higher, while the robustness of the Modified RRWEI-SM is stronger. In addition, due to exploiting the two-stage architecture, the Modified RRWEI-SM can be further improved by adopting the watermarking scheme that has better performance.

D. Theoretical Analysis

In this section, we analyze the embedding capacity and the security of the RRWEI-SM and the Modified RRWEI-SM. A derivation of the embedding capacity is given in the analysis of the embedding capacity. The security of the encrypted image is also proven.

1) *Analysis of the Embedding Capacity:* Because the PEE method is exploited in the RRWEI-SM, the smoothness of the image affects the maximum embedding capacity of the

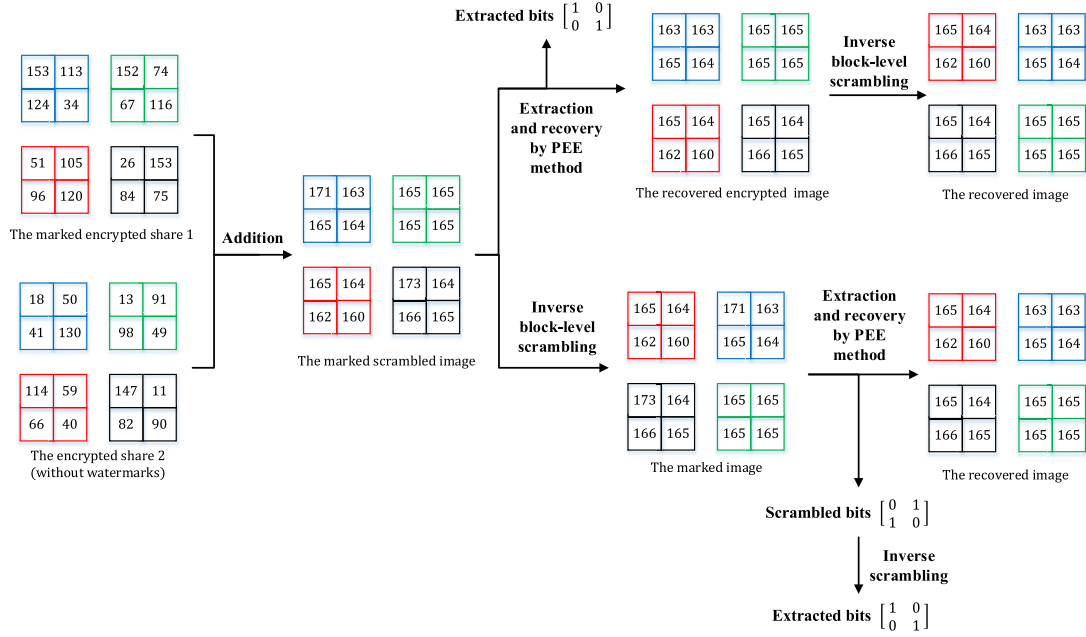


Fig. 8. An example of the extraction and recovery phase.

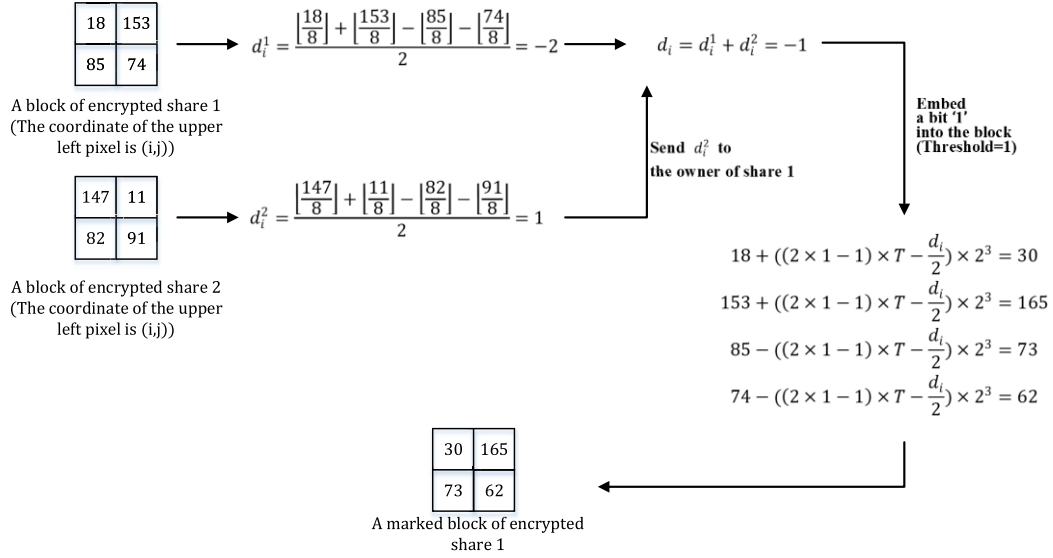


Fig. 9. An example of the robust embedding phase.

scheme. If a smooth cover image is used to embed the watermark bits, prediction values of the pixels in this image tend to be accurate. Distribution of the prediction errors will be centralized, which means that more bits can be embedded in one layer of the PEE. Suppose that k layers of the watermark are embedded into the share by the PEE. The size of the share is $M \times N$. In the i -th layer of the PEE, E_i represents the set of predictive errors, and M_{E_i} is calculated according to Eq. (20). For simplicity, only one round of embedding is performed in a layer of the PEE, i.e., each block of the cover image is used to embed one bit in a layer. The embedding capacity in this layer is equal to the number of occurrences when the predictive error values in the sequence E_i are equal to M_{E_i} . Thus, the

maximum embedding capacity can be obtained by

$$EC_{max} = \sum_{i=1}^k h_{E_i}(M_{E_i}) \quad (28)$$

where $h_{E_i}(X_i)$ is determined by the smoothness of the cover image. In PEE, the number of layers of the embedding is adjustable.

To ensure the visual quality of the marked image, the watermark of the PEE method is embedded into the cover image. The maximum embedding rate of the RRWEI-SM is $\frac{\sum_{i=1}^k h_{E_i}(M_{E_i})}{M \times N}$ bpp. Specifically, when the PSNR of the marked image in the RRWEI-SM scheme is set to 34 dB, the embedding capacities of the cover image 'Lena', 'Baboon' and

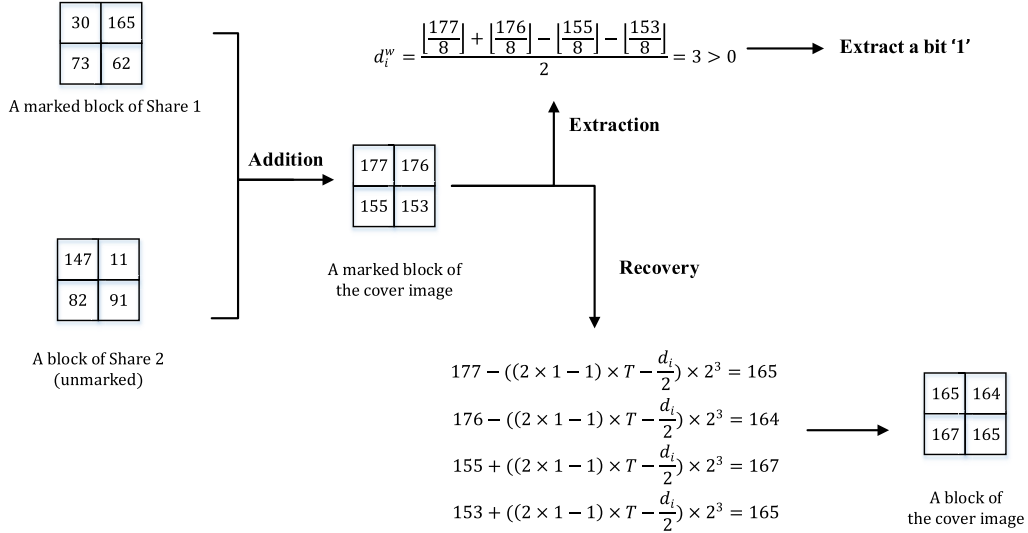


Fig. 10. An example of the extraction and recovery phase.

'Peppers' are approximately 0.6 *bpp*, 0.25 *bpp* and 0.55 *bpp*, respectively.

In the Modified RRWEI-SM scheme, the maximum embedding capacity can be reached by adjusting the size of the blocks in the patchwork robust watermarking. When a block consists of $2m(m > 0)$ pixels, the number of bits embedded into the cover image will be $\frac{M \times N}{2m}$. Because the minimum size of the patchwork block is 2×1 , the theoretical maximum capacity of the Modified RRWEI-SM is 0.5 *bpp*. The maximum capacity of other schemes will be analyzed in Section V.

2) *Analysis of Security:* We prove the security of the encryption method based on additive secret sharing and block-level scrambling. The analysis of the security includes three parts: the security of additive secret sharing, the security of the correlation of adjacent pixels and the security of the encrypted image.

Additive secret sharing splits the values of the pixels into two shares by addition randomly, and the security of the method can be proved by the following theorem.

Theorem 1: If a random element r is uniformly distributed on $\mathbb{Z}_n$ and is independent from any variable $x \in \mathbb{Z}_n$, then $r \pm x$ is also uniformly random and independent from x .

We refer to [39] for the proof of Theorem 1. Since $x_1(i, j)$ and $x_2(i, j)$ are uniformly random, the shares will not cause the leakage of the original pixel values. In the honest-but-curious model, the adversary is allowed to corrupt at most one of the two parties S_1 and S_2 . During the embedding phase, the two parties calculate part of $e_{HSB}(i, j)$ (i.e., $e_{HSB}^1(i, j)$ and $e_{HSB}^2(i, j)$, respectively). To prove that a protocol is secure, it suffices to show that the view of the corrupted party can be simulated given its input and output [34], [37].

For S_1 , the view in a calculation will be $view = (x_1(i, j), e_{HSB}^1(i, j), e_{HSB}^2(i, j), b)$. If S_1 tries to retrieve the value of $x(i, j)$ from $x_1(i, j)$, $x_1(i, j + 1)$, $x_1(i + 1, j)$, $x_1(i + 1, j + 1)$, $e_{HSB}^1(i, j)$ and $e_{HSB}^2(i, j)$ by prediction, then, (29), shown at the bottom of the next page. According to Theorem 1, it is trivial to see that the

retrieved value $x_p(i, j)$ is independent of the original pixel value $x(i, j)$. The reason is that $x_1(i, j)$, $x_1(i, j + 1)$, $x_1(i + 1, j)$ and $x_1(i + 1, j + 1)$ are parts of the shares of $x(i, j)$. Therefore, S_1 cannot recover $x(i, j)$ by $e_{HSB}^2(i, j)$ from S_2 .

In the Modified RRWEI-SM, two parts of the difference d_i are shared between the parties. The view of S_1 will be $(x_1(i, j), d_i^1, d_i^2, w)$. S_1 tries to retrieve the mean value of the pixel set in patchwork by

$$\overline{x_{\sigma(l)}} = \frac{\sum_{l=1}^m x_{\sigma(l)}}{m} \neq \frac{m \times (d_i^1 + d_i^2) + \sum_{l=m+1}^{2m} x_{\sigma(l)}}{m} \quad (30)$$

Since $x_{\sigma(l)}^1$ is uniformly random, the calculated average value $\overline{x_{\sigma(l)}}$ is independent from the original value.

The correlation between adjacent pixels in the image can be leaked. Therefore, block-level scrambling is applied in the encrypted method. The security depends on the selected high-speed-scrambling method [38]. Finally, the security of the encrypted image will be analyzed by the prevailing evaluation methods and shown in Section V-A. The results prove the security.

Generally, the lightweight image encryption method combined with additive secret sharing and block-level scrambling guarantees the security of our method to some degree.

V. EXPERIMENTS AND DISCUSSION

This section compares the performance and characteristics of our scheme with that of RRW schemes and reversible watermarking schemes in the encrypted image.

A. Experimental Results

We present the embedding rate, the visual quality of the marked image, the robustness of the watermark, and the security of the encrypted image. Comparisons between the proposed scheme and other schemes are shown.

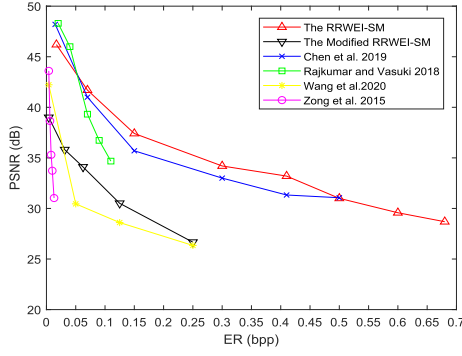


Fig. 11. Comparisons on average PSNR value of the three images.

1) *Embedding Rate and Visual Quality*: The embedding rate ER , which is used to evaluate the embedding capacity of the proposed scheme and other related schemes, is calculated by

$$ER = \frac{\text{number of watermark bits in the marked image}}{\text{number of pixels in the marked image}} (bpp) \quad (31)$$

In the decryption and extraction phase, if the marked encrypted image is first decrypted, the marked image will be obtained. The visual quality of the marked image is another evaluation criterion, and it is affected by the embedding rate and the number of HSBs. We compare the proposed schemes with some typical and state-of-the-art work [23], [26], [27], [32] in these terms. These schemes are the reversible watermarking in encrypted images or the robust reversible watermarking in the plaintext domain. By making a comparison with these schemes, we can obtain more comprehensive results.

Lena, Baboon and Peppers are used as the cover images in the experiment. The average values of the PSNR results of the three images are shown in Fig. 11, and it can be seen that the RRWEI-SM scheme has a higher visual quality compared with other schemes under 0.07 bpp or higher ER . Moreover, the maximum payload of the RRWEI-SM scheme is higher than that of the other schemes when the visual quality is above 30 dB. For example, Chen *et al.* [32] use difference expansion to embed one bit into a pair of pixels, and thus, the maximum payload is approximately 0.5 bpp . Wang *et al.* [23] use the patchwork watermarking to embed one bit into $2 \times n$ ($n > 0$) pixels, and the size of the cover image is half the size of the original image. Therefore, the maximum payload of Wang *et al.*'s scheme [23] is 0.25 bpp . The payload of the other schemes is limited, and the visual quality severely declines at a high payload, and thus, these schemes' results under the high embedding rate (above 0.5 bpp) are not shown. In the RRWEI-SM scheme, PEE can be performed on multiple layers and multiple rounds. As a result, the RRWEI-SM scheme achieves a high payload, while the visual quality decreases. When the embedding rate is

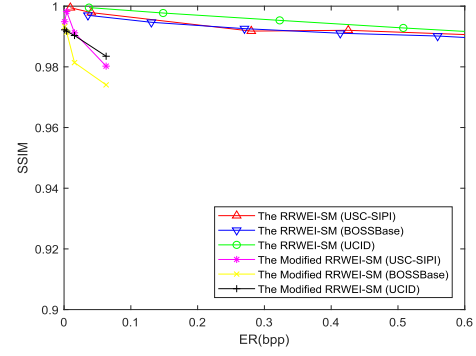


Fig. 12. SSIM value of the marked images in USC-SIPI, BOSSBase, and UCID.

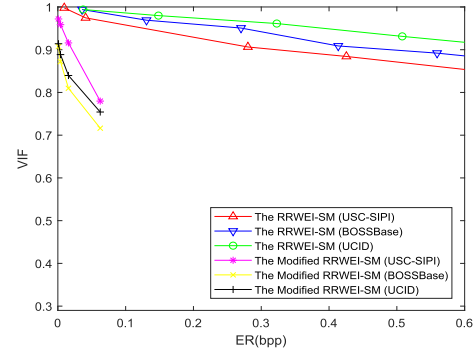


Fig. 13. VIF value of the marked images in USC-SIPI, BOSSBase, and UCID.

close to 0.7 bpp , the PSNR is higher than 30 dB for Lena and Peppers, and thus, we set it as the bound in the experiment. The visual quality of the Modified RRWEI-SM scheme is lower than that of RRWEI-SM. However, the watermark of the Modified RRWEI-SM scheme is more robust. The robustness experiments will be shown in Section V-A2.

We also use SSIM and VIF to evaluate the visual quality of the marked images with three different image datasets: USC-SIPI, BOSSBase, and UCID. When the value of SSIM and VIF is closer to 1, the visual quality of the marked image is higher. The experimental results are shown in Figs. 12 and 13. It can be seen that the visual quality of the marked image in RRWEI-SM is higher than that of the Modified RRWEI-SM. Moreover, in different datasets, the performance of the proposed schemes is stable. Thus, the proposed schemes are suitable for various applications.

2) *Robustness*: Many traditional RRW schemes evaluate the robustness of their watermarking at a low Embedding Capacity (EC). Thus, we will first analyze the robustness of the Modified RRWEI-SM scheme at an EC of 1024 bits and 256 bits. In addition, the RRWEI-SM and the Modified RRWEI-SM scheme can also ensure the robustness at a high EC. Therefore, we also evaluate the robustness of the watermark at the high EC.

$$x_p(i, j) = \left\lfloor \frac{x_{HSB}^1(i, j+1) + x_{HSB}^1(i+1, j) + x_{HSB}^1(i+1, j+1)}{3} \right\rfloor + \left\lfloor \frac{e_{HSB}^1(i, j) + e_{HSB}^2(i, j)}{3} \right\rfloor + x_{LSB}^1(i, j) \quad (29)$$

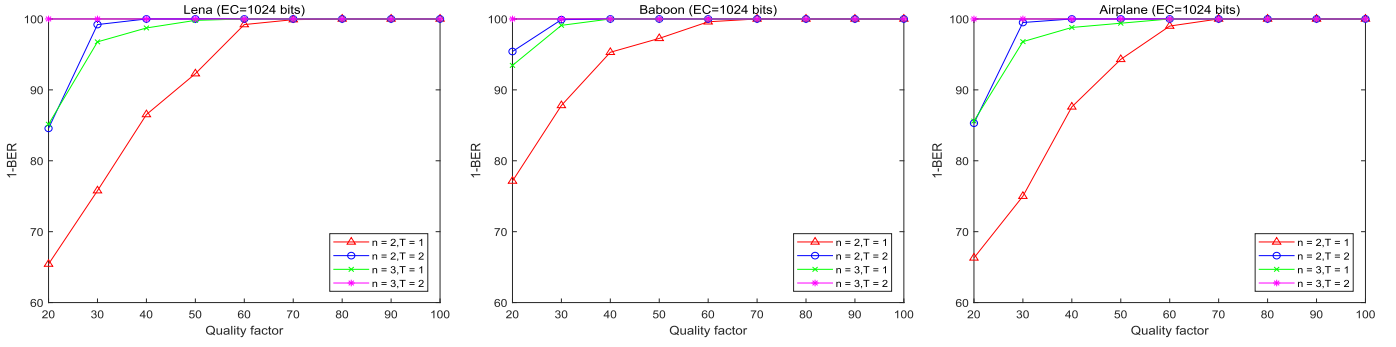


Fig. 14. Comparisons between different thresholds and HSB numbers in the Modified RRWEI-SM (when EC = 1024 bits).

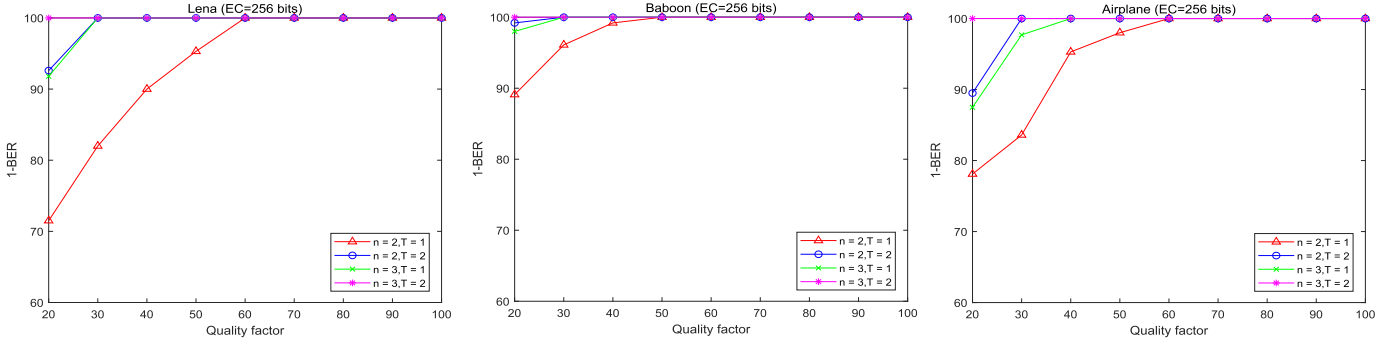


Fig. 15. Comparisons between different threshold and HSB numbers in the Modified RRWEI-SM (when EC = 256 bits).

We first evaluate the robustness of the watermarking's resistance to JPEG compression. The Bit Error Rate (BER) is utilized to indicate the robustness. The lower the BER value is, the better the robustness of the watermarking.

$$BER = \frac{\text{number of error watermark bits}}{\text{number of the original watermark bits}} \quad (32)$$

In the Modified RRWEI-SM scheme, the threshold of the patchwork robust watermarking and the number of the LSBs affect the robustness of the watermarking. The EC is set to 1024 bits and 256 bits, respectively, and the number of LSBs, n , is modified from 2 to 3. The threshold T of the patchwork watermarking is set to 1 and 2. The results of the robustness against JPEG compression with different quality factors are shown in Figs. 14 and 15. In these figures, the vertical axis denotes 1-BER. If the number of LSBs, n , and the threshold T , are increased, the robustness of the watermark will be enhanced, while the visual quality of the marked image will decline. As shown in Figs. 14 and 15, when $n = 3$ and $T = 2$, the watermark is the most robust.

In fact, the robustness is also affected by the embedding capacity. According to Eq. (25), more blocks are required when more watermark bits are embedded, and thus, the number of pixels in a block will decrease. The robustness of the watermark would be weakened. In other words, the robustness of the Modified RRWEI-SM scheme at EC = 256 bits is stronger than that at EC = 1024 bits.

Furthermore, we compare the robustness of the Modified RRWEI-SM scheme with that of some typical and state-of-the-art robust watermarking schemes [5], [19], [20], [22], [23].

When the PSNR of these schemes is adjusted to approximately 34 dB, the comparisons under different quality factors at an EC of 1024 bits are shown in Fig. 16. It can be seen that the accuracy of the extracted watermark in the Modified RRWEI-SM scheme is the highest among these schemes. When the embedding capacity equals 256 bits, the Modified RRWEI-SM can resist JPEG compression with a quality factor of 40, as shown in Fig. 17. The result is similar to Wang *et al.*'s scheme [23] and is better than the other schemes. In the method design, Wang *et al.*'s scheme also exploited the two-stage robust reversible watermarking in the frequency domain. However, Wang *et al.*'s scheme does not preserve the privacy of the cover image, and the maximum embedding capacity is lower than that of the proposed schemes. We also evaluate the robustness of the watermark's resistance to JPEG2000 compression with three different datasets. The experiment results are shown in Fig. 18. In this experiment, when $n = 3$ and $T = 2$, BER is the lowest, i.e., the watermark is the most robust. In Fig. 18, it can be seen that the performance of the Modified RRWEI-SM is stable under different image datasets. Moreover, if the quality of the compressed image is low (the compression ratio is equal to 50), more than 70% of the watermarks can still be extracted correctly at $n = 3$ and $T = 2$. Overall, the watermark of the Modified RRWEI-SM is robust.

The experiment results for resisting White Gaussian Noise (WGN) are shown in Table II. When the embedding capacity equals 256 bits, we set $n = 2$ and $T = 2$ to achieve a similar visual quality with the other schemes. As shown in Table II, the Modified RRWEI-SM scheme can resist WGN with 40 standard deviation. That is, better than An *et al.*'s and

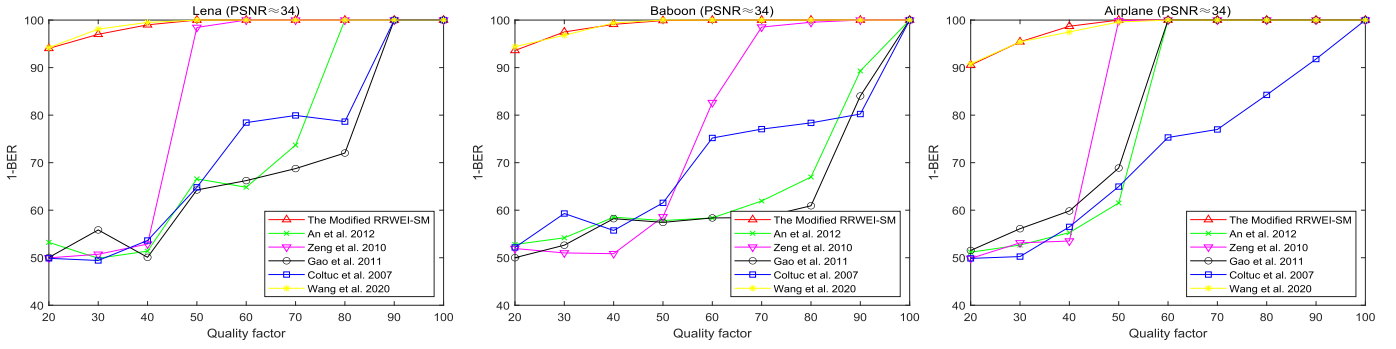


Fig. 16. Comparisons under different quality factors at EC of 1024 bits.

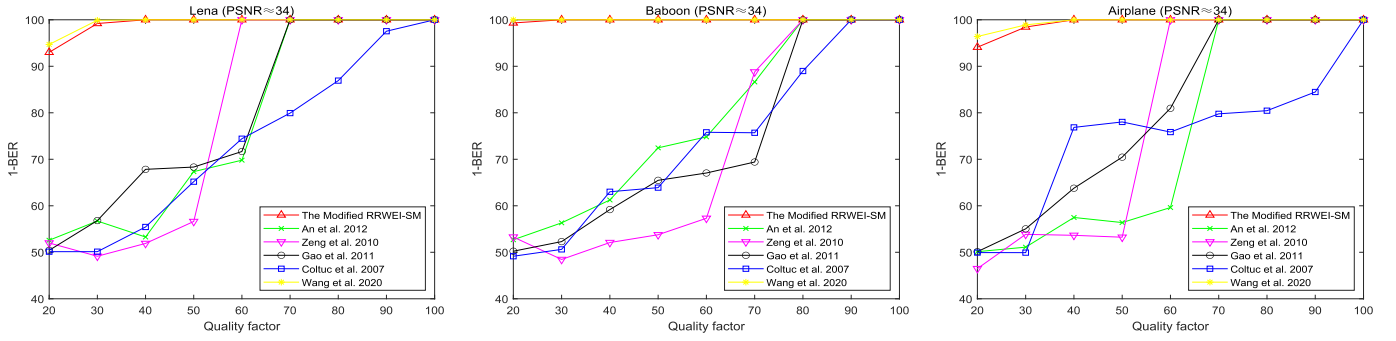


Fig. 17. Comparisons under different quality factors at EC of 256 bits.

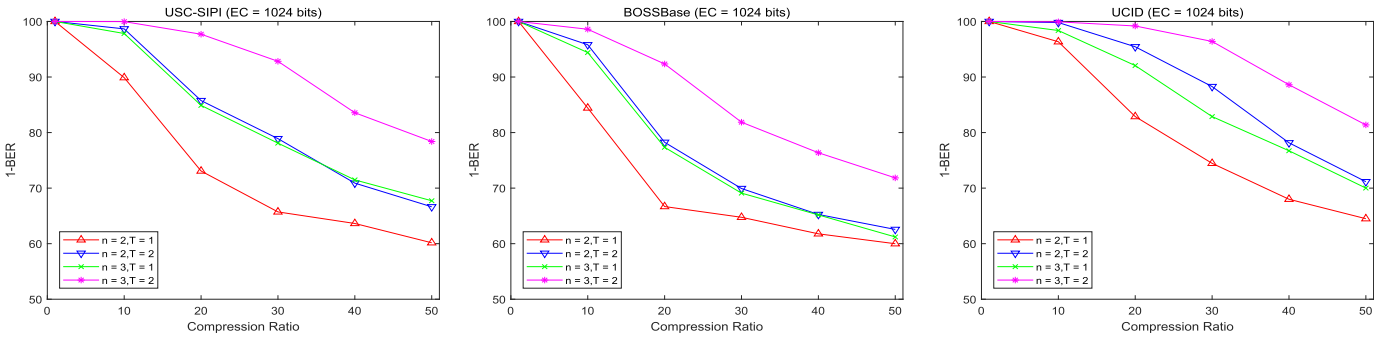


Fig. 18. Resisting JPEG2000 compression under different ratio at EC of 1024 bits.

Coltuc *et al.*'s schemes. The experiment results of resisting other noise and image filters are shown in Table III. It can be seen that robustness of the watermark increases with the threshold. Some attacked images and reconstructed images are demonstrated in Fig. 19. It can be seen that when the marked image is attacked, the cover image is distorted, but the BER of the extracted watermark is low in the Modified RRWEI-SM.

Compared with other schemes, the RRWEI-SM and the Modified RRWEI-SM scheme still have many other advantages, such as a higher maximum capacity. In [23], because each block carries at most one bit, the maximum capacity of Wang *et al.*'s scheme is 0.25 *bpp*, which is lower than that of the proposed schemes. To present the robustness of the RRWEI-SM scheme, the Modified RRWEI-SM and Wang *et al.*'s scheme at a high capacity of approximately 0.25 *bpp* (only consider the robustness), we take 0.25 *bpp* and 0.3 *bpp* as examples to compare with these three schemes.

TABLE II
COMPARISONS OF THE ROBUSTNESS AT THE EC
OF 256 BITS RESISTING WGN

Method	BER (resist WGN)			
	(0,10)	(0,20)	(0,30)	(0,40)
The Modified RRWEI-SM	0	0	0	0
Wang et al. 2020 [23]	0	0	0	0
An et al. 2012 [20]	0	10.2	11.5	12.0
Zeng et al. 2010 [19]	0	0	0	0
Gao et al. 2011 [5]	0	0	0	0
Coltuc et al. 2007 [22]	32.7	32.4	32.6	32.6

The BER results for resisting WGN and JPEG compression are presented in Table IV. The PSNR of the marked image

TABLE III
EXPERIMENT OF RESISTING ATTACKS

Threshold of the Modified RRWEI-SM			BER			
			n=2, T=1	n=2, T=2	n=3, T=1	n=3, T=2
Image Filter	Gaussian Filter	3×3	14.26	5.28	5.67	1.38
		5×5	16.90	8.44	7.91	3.13
		7×7	17.09	8.64	8.69	3.03
	Median Filter	3×3	12.49	4.59	4.50	0.79
		5×5	24.88	16.12	16.51	7.82
		7×7	34.38	25.88	25.91	17.29
Noise	WGN	(0,20)	0.87	0	0	0
		(0,40)	6.82	0.20	0.69	0
		(0,60)	14.28	2.06	3.52	0
	Salt & Pepper	0.1	9.87	0.59	1.37	0
		0.2	17.97	3.33	4.30	0
		0.4	30.28	16.51	17.09	3.13
Poisson		0	0	0	0	

TABLE IV
COMPARISONS OF THE ROBUSTNESS AT THE HIGH EC

Method	ER (bpp)	BER (resist WGN)				BER (resist JPEG compression)				PSNR
		(0,1)	(0,2)	(0,5)	(0,10)	q=90	q=80	q=70	q=60	
The RRWEI-SM	0.25	11.5	22.7	42.2	48.3	34.6	42.2	45.3	47.1	36.17
	0.3	14.5	24.2	42.8	47.9	35.8	43.7	46.6	47.5	35.36
The Modified RRWEI-SM	0.25	0.5	0.7	7.3	21.8	4.4	18.5	26.7	31.8	30.03
	0.3	2.3	3.0	10.9	25.4	11.2	22.5	30.1	35.1	29.24
Wang et al. 2020 [23]	0.25	4.6	19.5	39.8	46.4	31.4	42.3	46.1	47.9	29.05
	0.3	-	-	-	-	-	-	-	-	-

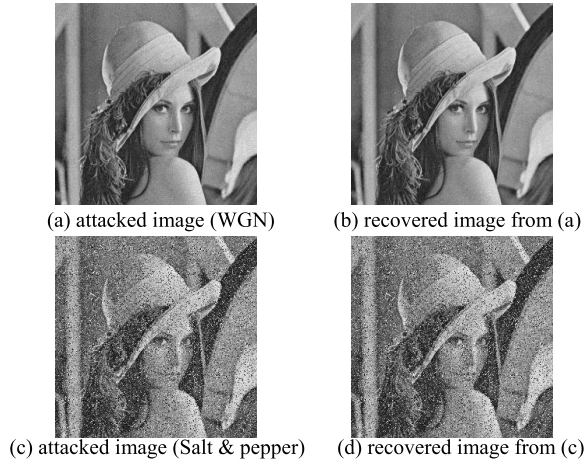


Fig. 19. Attacked images and the recovered images. (a) PSNR = 27.36 (b) PSNR = 28.13, BER = 0 (c) PSNR = 12.34 (d) PSNR = 12.45, BER = 0.033.

in RRWEI-SM is the highest. In addition, the PSNR of the marked image in the Modified RRWEI-SM is the second highest. At an EC of 0.25 *bpp*, the robustness of the RRWEI-SM scheme is close to Wang *et al.*'s scheme. The Modified RRWEI-SM has the best robustness. When the EC equals 0.3 *bpp*, the watermarking robustness of the two proposed schemes is stable and the EC of Wang *et al.*'s scheme can only reach 0.25 *bpp*. Overall, compared with Wang *et al.*'s scheme, the RRWEI-SM scheme and the Modified RRWEI-SM scheme

TABLE V
NPCR AND CORRELATION FOR VARIOUS BLOCK SIZES

Block Size	NPCR	Correlation		
		<i>H</i>	<i>V</i>	<i>D</i>
2×2	98.72%	0.1526	0.1148	0.0674
4×4	98.68%	0.1987	0.2022	0.1535
8×8	98.55%	0.2308	0.2317	0.1971
16×16	98.41%	0.2570	0.2536	0.2282

Note: *H*, *V* and *D* denote the horizontal, vertical and diagonal directions.

have better visual quality of the marked image at high EC. The robustness of the watermark in the Modified RRWEI-SM is also high.

3) *Security of the Encrypted Image*: To evaluate the security of the encrypted image, the ability to resist differential attacks and the correlation between adjacent pixels of the encrypted image will be analyzed [38].

The ability to resist differential attacks can be quantitatively tested by the Number of Pixel Change Rate (NPCR).

$$NPCR = \frac{\sum_{i,j} D(i,j)}{M \times N} \times 100\% \quad (33)$$

where

$$D(i,j) = \begin{cases} 1, & \text{if } C_1(i,j) \neq C_2(i,j) \\ 0, & \text{if } C_1(i,j) = C_2(i,j) \end{cases} \quad (34)$$

$C_1(i,j)$ and $C_2(i,j)$ ($i = 1, 2, \dots, M, j = 1, 2, \dots, N$) denote the encrypted images before and after one bit of

TABLE VI
COMPARISONS OF THE CHARACTERISTICS WITH OTHER SCHEMES

Method	Characteristics					
	Robustness	Reversibility	Privacy preserving	Encryption method	Separable	Multi-party watermarking
The RRWEI-SM	Yes	Yes	Yes	Additive secret sharing and scrambling	Yes	Yes
The Modified RRWEI-SM	Yes	Yes	Yes	Additive secret sharing and scrambling	Yes	Yes

the plain image has changed. The ideal value of NPCR is 99.6094% for a 256-gray-level image. If NPCR is closer to the ideal value, then the security of the encryption method is higher.

We experiment on the security of the encrypted image using 10 random images in three datasets, including USC-SIPI, BOSSBase, and UCID. The average results are shown in Table V. When the block-size of the encrypted image is 2×2 , the NPCR is 98.72%. The NPCR with different block-sizes 4×4 , 8×8 and 16×16 are 98.68%, 98.55% and 98.41%, respectively. Therefore, the ability to resist differential attack is guaranteed.

The adjacent pixels in the plain image are strongly correlated in the horizontal, vertical and diagonal directions. Attackers would glean some information on the cover image from the correlation of the pixels [40]. Therefore, the correlation must be analyzed in the encrypted image. We evaluate the correlation between the pixels of the encrypted image. 5000 pairs of pixels in the horizontal, vertical, and diagonal directions are selected, and the correlation coefficients are calculated by

$$\begin{cases} E(x) = \frac{1}{p} \sum_{i=1}^p x_i, \\ D(x) = \frac{1}{p} \sum_{i=1}^p (x_i - E(x))^2, \\ cov(x, y) = \frac{1}{p} \sum_{i=1}^p (x_i - E(x))(y_i - E(y)), \\ r_{xy} = \frac{cov(x, y)}{\sqrt{D(x)} \times \sqrt{D(y)}} \end{cases} \quad (35)$$

where x and y are the selected pixel sequences. Here, $P = 5000$. $E(x)$ and $D(x)$ represent the expectation and variance of x , respectively.

The encryption method should reduce the correlation between the pixels. The correlation value is closer to zero, and the security of the encrypted image is higher. As shown in Table V, the smaller the block size is, the weaker the correlation between the adjacent pixels remains. The block-size 2×2 has the best performance. Therefore, the security of the encrypted image is proven.

B. Comparisons and Discussion

The RRWEI-SM and the Modified RRWEI-SM scheme possess the characteristics of robust reversible watermarking in encrypted images (i.e., the robustness of the watermarking, the reversibility of the cover image, and privacy preserving of the cover image). The watermark of the proposed schemes is robust, and the cover image can be recovered. Additionally,

TABLE VII
COMPUTATION COMPLEXITY

Scheme	Encryption complexity	Decryption complexity
Xiang <i>et al.</i> 2018 [31]	$O(n^3)$	$O(n^3)$
Chen <i>et al.</i> 2019 [32]	$O(n)$	$O(n \log^2 n)$
Wu <i>et al.</i> 2018 [8]	$O(n)$	$O(n \log^2 n)$
Ke <i>et al.</i> 2020 [17]	$O(n^2)$	$O(n^2)$
Puteaux and Puech 2018 [30]	$O(n)$	$O(n)$
The RRWEI-SM	$O(n)$	$O(n)$
The Modified RRWEI-SM	$O(n)$	$O(n)$

the privacy of the cover image is preserved with the encryption method based on additive secret sharing and block-level scrambling. The characteristics of the proposed schemes are shown in Table VI. The comparison between other schemes has been presented in Table I.

1) *Characteristics*: The characteristics of the other schemes are as follows. Wang *et al.* [23], Wang *et al.* [24], and Kumar and Jung [25] have robustness and reversibility, but the privacy of the cover image cannot be preserved. Xiang *et al.* [31] and Ke *et al.* [17] utilize homomorphic encryption to protect the privacy of the cover image. However, robustness of the embedded data is not achieved, and the computational complexity is high. Puteaux and Puech [30] use a stream cipher to encrypt the cover image, but the scheme is not separable. Chen *et al.* [32] and Wu *et al.* [8] propose the RDHEI, which exploits secret image sharing to reduce the computational complexity. However, they did not achieve the robustness of the secret message. Moreover, these schemes involve only one data hider, and thus, they are not suitable for multi-party watermarking applications.

The RRWEI-SM and the Modified RRWEI-SM scheme develop additive secret sharing and block-level scrambling as a lightweight encryption method. Moreover, a robust reversible watermark can be embedded into the encrypted images. The proposed schemes are separable and are suitable for multi-party watermarking. In this way, the RRWEI-SM and the Modified RRWEI-SM scheme can be applied to more various applications scenarios.

2) *Computational Complexity*: We compare the computational complexity of the proposed encryption method with that of other schemes. The comparison results are presented in Table VII. The homomorphic encryption-based schemes

[17], [31] are separable, but the encryption and decryption complexity is high. Chen *et al.* [32] and Wu *et al.* [8] utilize secret image sharing to encrypt the cover image. The complexity of the polynomial-based encryption is $O(n)$, and the Lagrange interpolation operation-based decryption's complexity is $O(n \log^2 n)$. The stream cipher-based scheme [30] has a low computational complexity, but the scheme is not separable. The proposed schemes develop additive secret sharing and block-level scrambling, and thus, the complexity of the encryption and decryption is $O(n)$. Overall, the proposed schemes are secure, robust and have a low computational complexity.

VI. CONCLUSION

In this paper, we make the following conclusions with regard to the main issues that exist in current reversible watermarking schemes: 1) The embedding capacity of traditional RRW is limited; 2) The RRW in the plaintext domain results in the leakage of privacy; 3) The complexity of the encryption method for the cover image is relatively high; 4) Inseparable schemes would reduce the practicality of robust reversible watermarking; 5) Current RRW schemes are not suitable for multiparty copyright protection applications.

To solve these issues, this paper proposes the RRWEI-SM and the Modified RRWEI-SM scheme. Additive secret sharing and block-level scrambling are developed to encrypt the image. In the RRWEI-SM, reversible watermarking is embedded into the HSB plane of the cover image by the PEE method. In the Modified RRWEI-SM, based on the two-stage architecture, the patchwork robust watermarking and PEE-based reversible watermarking are utilized to embed the watermark. Multiple parties are involved in the embedding phase to calculate the characteristics of the shares cooperatively with no privacy leakage. The cover image can be recovered after the watermark extraction. Experimental results demonstrate that the performance of the RRWEI-SM scheme and the Modified RRWEI-SM scheme are superior to that of traditional RW methods. In the RRWEI-SM, the visual quality of the marked image and the embedding capacity are higher. It yields a close robustness with several state-of-the-art RRW methods at a high EC. The watermark of the Modified RRWEI-SM is more robust at a low EC. This finding means that the two proposed schemes are suitable for different applications.

The proposed schemes still have some drawbacks. The robustness of the watermark in the RRWEI-SM scheme is based on the property of HSBs, and it must be improved. The modified RRWEI-SM achieves a high robustness of the watermark, but the visual quality of the marked image is not high. Moreover, the security of the encryption method based on additive secret sharing and block-level scrambling should be further enhanced.

To obtain a different performance for the proposed scheme, many other RRW techniques can also be introduced to the framework of the RRWEI-SM. In future work, we plan to study new RRW methods to improve the robustness of the watermark and the visual quality of the marked image under different embedding rates. At the same time, the lightweight encryption method is further studied by other secret sharing

methods (e.g., bitwise XOR) to enhance the security of the encrypted image.

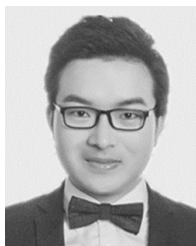
ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valued comments which helped to improve this article.

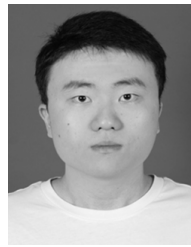
REFERENCES

- [1] Y.-Q. Shi, X. Li, X. Zhang, H.-T. Wu, and B. Ma, "Reversible data hiding: Advances in the past two decades," *IEEE Access*, vol. 4, pp. 3210–3237, 2016.
- [2] M. Goljan, J. Fridrich, and R. Du, "Distortion-free data embedding for images," in *Proc. Int. Workshop Inf. Hiding*. Berlin, Germany: Springer, 2001, pp. 27–41.
- [3] M. U. Celik, G. Sharma, A. M. Tekalp, and E. Saber, "Lossless generalized-LSB data embedding," *IEEE Trans. Image Process.*, vol. 14, no. 2, pp. 253–266, Feb. 2005.
- [4] Z. Ni, Y.-Q. Shi, N. Ansari, and W. Su, "Reversible data hiding," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 16, no. 3, pp. 354–362, Mar. 2006.
- [5] X. Gao, L. An, Y. Yuan, D. Tao, and X. Li, "Lossless data embedding using generalized statistical quantity histogram," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 21, no. 8, pp. 1061–1070, Aug. 2011.
- [6] C. Dragoi and D. Coltuc, "Improved rhombus interpolation for reversible watermarking by difference expansion," in *Proc. 20th Eur. Signal Process. Conf. (EUSIPCO)*, Aug. 2012, pp. 1688–1692.
- [7] X. Li, B. Li, B. Yang, and T. Zeng, "General framework to histogram-shifting-based reversible data hiding," *IEEE Trans. Image Process.*, vol. 22, no. 6, pp. 2181–2191, Jun. 2013.
- [8] X. Wu, J. Weng, and W. Yan, "Adopting secret sharing for reversible data hiding in encrypted images," *Signal Process.*, vol. 143, pp. 269–281, Feb. 2018.
- [9] J. Wang, J. Ni, X. Zhang, and Y.-Q. Shi, "Rate and distortion optimization for reversible data hiding using multiple histogram shifting," *IEEE Trans. Cybern.*, vol. 47, no. 2, pp. 315–326, Feb. 2017.
- [10] J. Tian, "Reversible data embedding using a difference expansion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 13, no. 8, pp. 890–896, Aug. 2003.
- [11] I.-C. Dragoi and D. Coltuc, "Local-prediction-based difference expansion reversible watermarking," *IEEE Trans. Image Process.*, vol. 23, no. 4, pp. 1779–1790, Apr. 2014.
- [12] S. Weng, Y. Chen, W. Hong, J.-S. Pan, C.-C. Chang, and Y. Liu, "An improved integer transform combining with an irregular block partition," *Symmetry*, vol. 11, no. 1, p. 49, Jan. 2019.
- [13] D. Coltuc, "Improved embedding for prediction-based reversible watermarking," *IEEE Trans. Inf. Forensics Security*, vol. 6, no. 3, pp. 873–882, Sep. 2011.
- [14] X. Li, B. Yang, and T. Zeng, "Efficient reversible watermarking based on adaptive prediction-error expansion and pixel selection," *IEEE Trans. Image Process.*, vol. 20, no. 12, pp. 3524–3533, Dec. 2011.
- [15] I.-C. Dragoi and D. Coltuc, "On local prediction based reversible watermarking," *IEEE Trans. Image Process.*, vol. 24, no. 4, pp. 1244–1246, Apr. 2015.
- [16] I.-C. Dragoi and D. Coltuc, "Adaptive pairing reversible watermarking," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2420–2422, May 2016.
- [17] Y. Ke, M.-Q. Zhang, J. Liu, T.-T. Su, and X.-Y. Yang, "Fully homomorphic encryption encapsulated difference expansion for reversible data hiding in encrypted domain," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 8, pp. 2353–2365, Aug. 2020.
- [18] L. Xiong and D. Dong, "Reversible data hiding in encrypted images with somewhat homomorphic encryption based on sorting block-level prediction-error expansion," *J. Inf. Secur. Appl.*, vol. 47, pp. 78–85, Aug. 2019.
- [19] X.-T. Zeng, L.-D. Ping, and X.-Z. Pan, "A lossless robust data hiding scheme," *Pattern Recognit.*, vol. 43, no. 4, pp. 1656–1667, Apr. 2010.
- [20] L. An, X. Gao, X. Li, D. Tao, C. Deng, and J. Li, "Robust reversible watermarking via clustering and enhanced pixel-wise masking," *IEEE Trans. Image Process.*, vol. 21, no. 8, pp. 3598–3611, Aug. 2012.
- [21] D. Coltuc, "Towards distortion-free robust image authentication," *J. Phys., Conf. Ser.*, vol. 77, Jul. 2007, Art. no. 012005.
- [22] D. Coltuc and J.-M. Chassery, "Distortion-free robust watermarking: A case study," *Proc. SPIE*, vol. 6505, Feb. 2007, Art. no. 65051N.

- [23] X. Wang, X. Li, and Q. Pei, "Independent embedding domain based two-stage robust reversible watermarking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 8, pp. 2406–2417, Aug. 2020.
- [24] W. Wang, J. Ye, T. Wang, and W. Wang, "Reversible data hiding scheme based on significant-bit-difference expansion," *IET Image Process.*, vol. 11, no. 11, pp. 1002–1014, Nov. 2017.
- [25] R. Kumar and K.-H. Jung, "Robust reversible data hiding scheme based on two-layer embedding strategy," *Inf. Sci.*, vol. 512, pp. 96–107, Feb. 2020.
- [26] R. Rajkumar and A. Vasuki, "Reversible and robust image watermarking based on histogram shifting," *Cluster Comput.*, vol. 22, no. S5, pp. 12313–12323, Sep. 2019.
- [27] T. Zong, Y. Xiang, I. Natgunanathan, S. Guo, W. Zhou, and G. Beliakov, "Robust histogram shape-based method for image watermarking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 5, pp. 717–729, May 2015.
- [28] N. Memon and P. Wah Wong, "A buyer-seller watermarking protocol," *IEEE Trans. Image Process.*, vol. 10, no. 4, pp. 643–649, Apr. 2001.
- [29] F. Frattolillo, "A multiparty watermarking protocol for cloud environments," *J. Inf. Secur. Appl.*, vol. 47, pp. 246–257, Aug. 2019.
- [30] P. Puteaux and W. Puech, "An efficient MSB prediction-based method for high-capacity reversible data hiding in encrypted images," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 7, pp. 1670–1681, Jul. 2018.
- [31] S. Xiang and X. Luo, "Reversible data hiding in homomorphic encrypted domain by mirroring ciphertext group," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 11, pp. 3099–3110, Nov. 2018.
- [32] Y.-C. Chen, T.-H. Hung, S.-H. Hsieh, and C.-W. Shiu, "A new reversible data hiding in encrypted image based on multi-secret sharing and lightweight cryptographic algorithms," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 12, pp. 3332–3343, Dec. 2019.
- [33] X. Zhang, J. Long, Z. Wang, and H. Cheng, "Lossless and reversible data hiding in encrypted images with public-key cryptography," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 26, no. 9, pp. 1622–1631, Sep. 2016.
- [34] K. Huang, X. Liu, S. Fu, D. Guo, and M. Xu, "A lightweight privacy-preserving CNN feature extraction framework for mobile sensing," *IEEE Trans. Depend. Sec. Comput.*, early access, Apr. 26, 2020, doi: 10.1109/TDSC.2019.2913362.
- [35] A. Shamir, "How to share a secret," *Commun. ACM*, vol. 22, no. 11, pp. 612–613, Nov. 1979.
- [36] G. R. Blakley, "Safeguarding cryptographic keys," in *Proc. Int. Workshop Manag. Requirements Knowl. (MARK)*, Jun. 1979, pp. 313–317.
- [37] D. Bogdanov, S. Laur, and J. Willemson, "Sharemind: A framework for fast privacy-preserving computations," in *Proc. Eur. Symp. Res. Comput. Secur.* Berlin, Germany: Springer, 2008, pp. 192–206.
- [38] Z. Hua, S. Yi, and Y. Zhou, "Medical image encryption using high-speed scrambling and pixel adaptive diffusion," *Signal Process.*, vol. 144, pp. 134–144, Mar. 2018.
- [39] D. Bogdanov, M. Niitsoo, T. Toft, and J. Willemson, "High-performance secure multi-party computation for data mining applications," *Int. J. Inf. Secur.*, vol. 11, no. 6, pp. 403–418, Nov. 2012.
- [40] Y. Chen, C. Tang, and R. Ye, "Cryptanalysis and improvement of medical image encryption using high-speed scrambling and pixel adaptive diffusion," *Signal Process.*, vol. 167, Feb. 2020, Art. no. 107286.



Lizhi Xiong received the Ph.D. degree in communication and information system from Wuhan University, China, in 2016. From 2014 to 2015, he was the Joint Ph.D. student with the Department of Electrical and Computer Engineering, New Jersey University of Technology, Newark, NJ, USA. He is currently an Associate Professor with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing, China. His main research interests include information hiding, encrypted image processing, and cloud computing security.



Xiao Han received the B.S. degree in software engineering from the School of Computer and Software, Nanjing University of Information Science and Technology, China, in 2020, where he is currently pursuing the master's degree in computer science and technology. His research interests include reversible data hiding in encrypted images and multimedia processing.



Ching-Nung Yang (Senior Member, IEEE) received the B.S. and M.S. degrees in telecommunication engineering from National Chiao Tung University, Hsinchu, Taiwan, in 1983 and 1985, respectively, and the Ph.D. degree in electrical engineering from National Cheng Kung University, Tainan, Taiwan, in 1997. He is currently a Full Professor with the Department of Computer Science and Information Engineering, National Dong Hwa University, Hualien, Taiwan. His research interests include coding theory, information security, and cryptography. He is a fellow of IET.



Yun-Qing Shi (Life Fellow, IEEE) received the B.S. and M.S. degrees from Shanghai Jiao Tong University, Shanghai, China, and the M.S. and Ph.D. degrees from the University of Pittsburgh, Pittsburgh, PA, USA. He has been a Professor with the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, NJ, USA, since 1987. He has authored and coauthored more than 300 articles, one book, and five book chapters, and has edited ten books. He holds 28 U.S. patents. His current research interests include digital data hiding, forensics and information assurance, and visual signal processing and communications. He is a member of a few IEEE Technical Committees. He received the Innovators Award 2010 from the New Jersey Inventors Hall of Fame for Innovations in Digital Forensics and Security. His U.S. patent 7,457,341 titled "System and Method for Robust Reversible Data Hiding and Data Recovery in the Spatial Domain" received the 2010 Thomas Alva Edison Patent Award from the Research and Development Council of New Jersey. He served as an Associate Editor for the IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, IEEE TRANSACTIONS ON SIGNAL PROCESSING, and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS, and a few other journals, the Technical Program Chair for the 2007 IEEE International Conference on Multimedia and Expo, a Co-Technical Chair for the 2006, 2007, and 2009–2015 International Workshop on Digital Watermarking, and the 2005 IEEE International Workshop on Multimedia Signal Processing (MMSP), a Co-General Chair for the 2002 IEEE MMSP, and a Distinguished Lecturer for the IEEE Circuits and Systems Society.